A financed-emissions number is its score distribution

The estimand

Scope 1 financed emissions, in tCO2e, of a loan book to US corporates under the PCAF business-loan attribution (outstanding amount over equity plus debt), with the PCAF data-quality score on every line. The borrowers, their emissions and their balance sheets are real. The bank is simulated: which borrowers it lends to, and how much.

The answer

100 loans, USD 37,957 mn outstanding, 10.12 Mt financed
[9.24 Mt, 16.12 Mt], 267 tCO2e per USD mn lent. Weighted
data-quality score 2.39: 80% of the financed emissions sit on reported lines.

The same book with no reported data (score 4.04): 13.61 Mt, and the realised-error interval is [5.33 Mt, 21.41 Mt], 4.0 times wide end to end.
Reported emissions for this book total 10.27 Mt. Both estimates cover it.

Why the score distribution is the result

The headline alone cannot say which of the two books it came from. The first estimate is within 1.4% of the reported figure; the second is +33% off.
Same borrowers, same attribution, same estimand. The score tells the reader which number to trust, so it belongs on the same line as the total.

Method and data

Data, all fetched by code (accessed 2026-10-12)

Emissions: EPA GHGRP facility table and emissions facts, data.epa.gov/efservice, direct emitters only, with the parent-company ownership shares. 11,281 facilities in 2023.
Financials and sector: SEC EDGAR XBRL company facts and submissions (revenue, assets, equity, debt, SIC code). Join: exact normalised name against SEC's ticker list.
257 parents matched; 210 usable in 2023 after dropping missing or non-positive fields.
Matched parents hold 37% of all reported US direct emissions in 2023.

Attribution and score

Financed emissions of a line = outstanding / (equity + debt) x borrower emissions.
Score 2: reported emissions (GHGRP, unverified by a third party). Score 4: sector emission factor x revenue. Score 5: sector factor x assets. No score 1 or 3 exists in public data. The factor is the sector's total emissions over its total revenue or assets, from borrowers that are not in the book. Sector: SIC 2-digit, pooled if fewer than 5 peers. Rule: the largest 40% of the book by revenue report, the next 40% use revenue, the rest use assets. Loans are 1-5% of the borrower's debt, drawn at random.

Uncertainty

Each sector's peer group is resampled once per replicate and sets the factor for every line in that sector. Each line then draws its own departure from the sector average. Reported lines carry none. The book is the population being estimated and is not resampled. Because borrowers are real, the estimate is also tested: 300 random books, factor fitted on the borrowers outside the book, compared to the reported total.

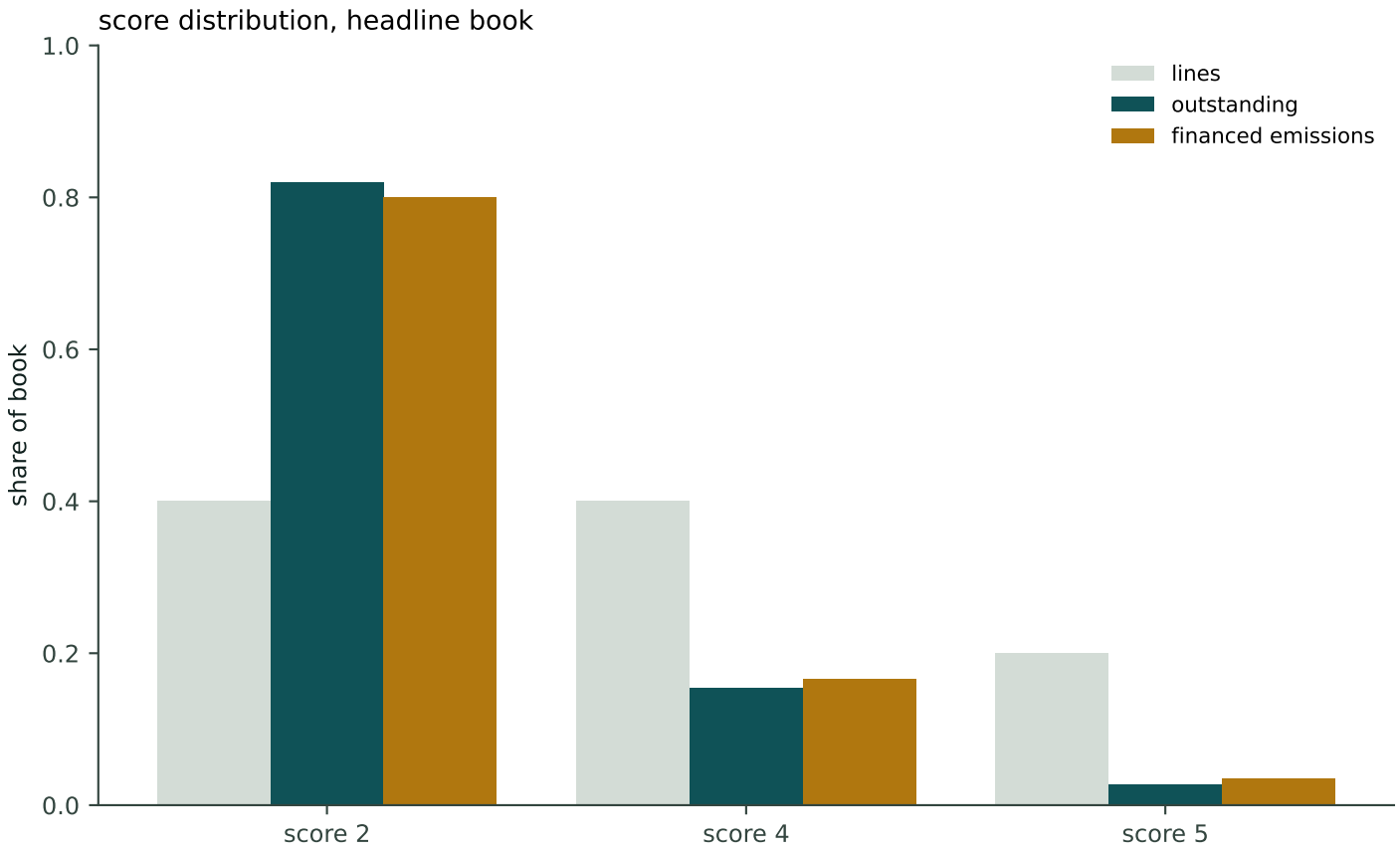
Result: score distribution beside the total

The book

Score 2 lines are 40 of 100 and 82% of the money, because large firms report.
Score 4 and 5 lines are 60 of 100 and 20% of the financed emissions.
The five largest lines hold 52% of the total; 5 of them are electric and gas utilities (SIC 49) and 5 are reported lines, so they decide the number and carry no estimate error.

Where the interval comes from

Randomising only score 4 lines gives a width of 4.75 Mt, only score 5 4.88 Mt, both 6.88 Mt.
Score 5 lines are 20% of the lines and 2.6% of the money; their width is 1.03 times that of the score 4 tier, which holds 15% of the money.



Validation: estimates against what was reported

Line level

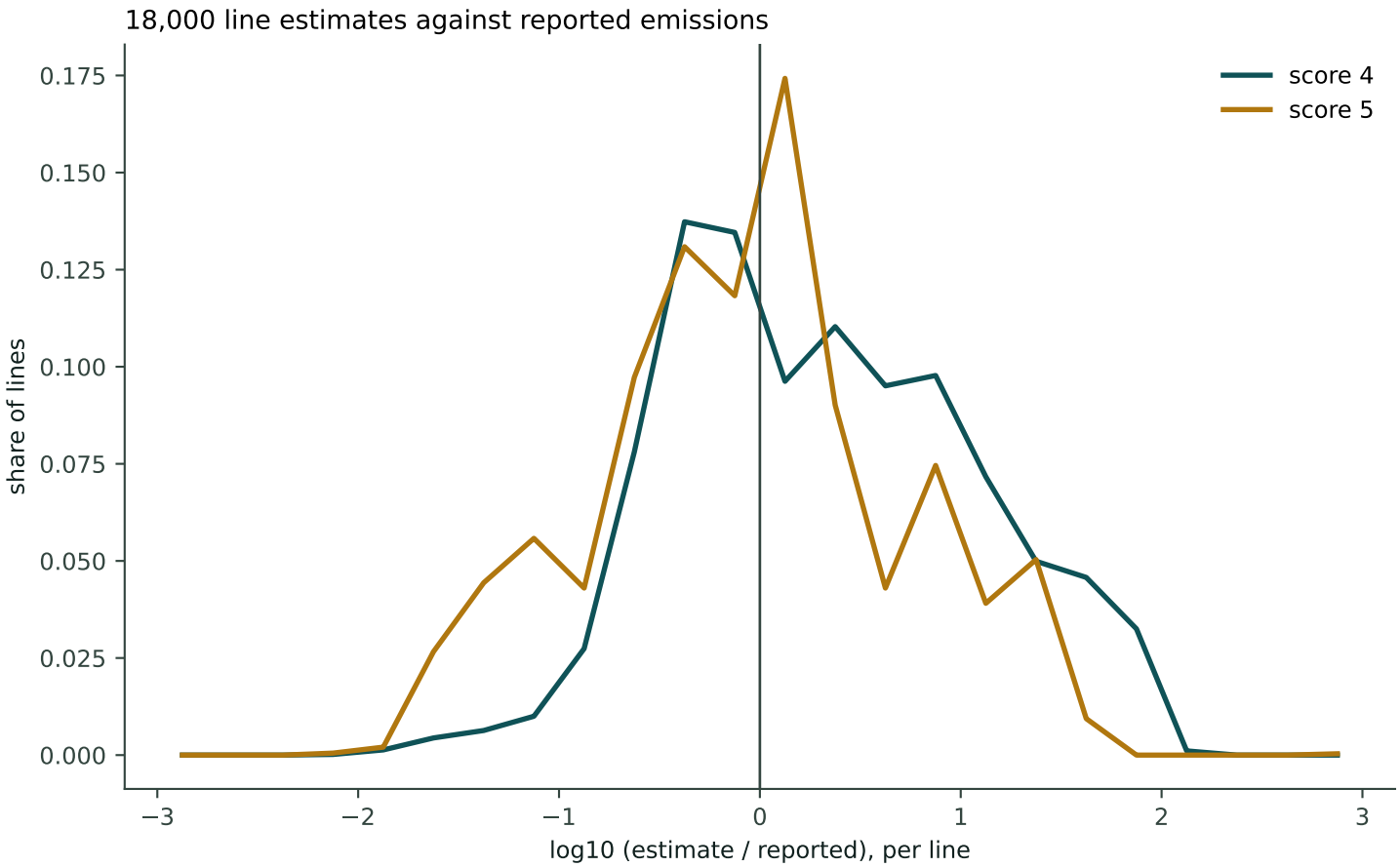
Score 4: median estimate is 1.83 times the reported figure, 28% of lines within a factor of 2, 78% within a factor of 10, 90% range 0.18 to 46.
Score 5: median 0.98, 34% within 2, 75% within 10, 90% range 0.04 to 26.

Book level, 300 random books

Headline mix: the interval covers the reported total in 97% of books (nominal 95%).
Estimate over reported: median 0.94, 90% range 0.83 to 1.08.
No reported data: coverage 64%. The truth fell below the interval in 109 books.
Cause: within a sector, intensity falls with size (slope -0.32 on log revenue), the sector average ignores it, and the largest lines drive the sum. The peer resample cannot see a bias common to all peers, so for this case the interval is taken from the realised error: [5.33 Mt, 21.41 Mt].

What this does not validate

Reported emissions are the reference, not a second measurement. No independent public source measures the financed emissions of a simulated book.



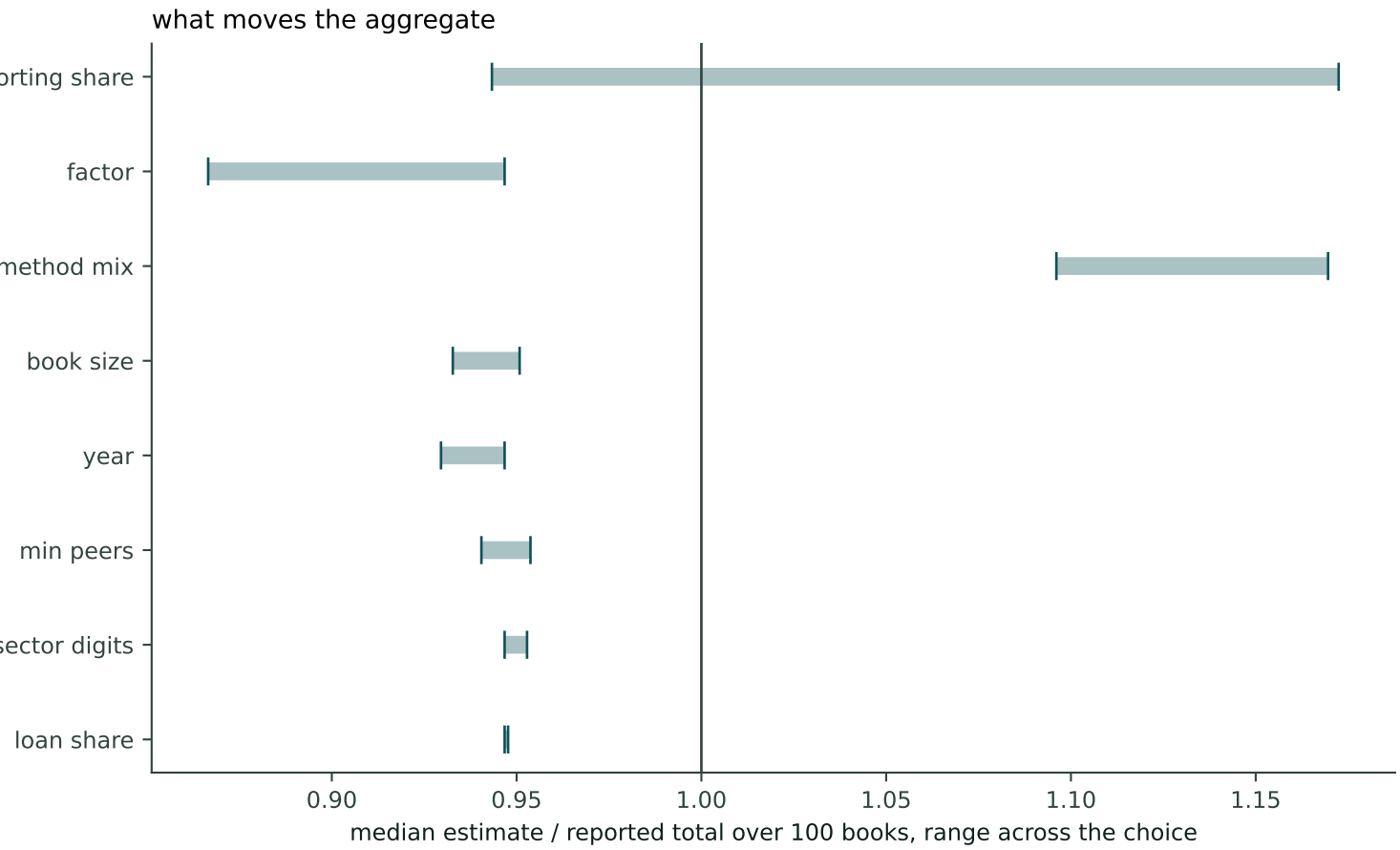
Sensitivity: the share of reported lines decides

Aggregate

Largest swing in the median estimate over reported: reporting share, 0.23 (0.94 to 1.17). Next: factor, 0.08.
Sector granularity moves it by 0.006, the reference year by 0.017, the loan size by 0.001. Book size 0.018, minimum peers 0.013.

Line level

Share of estimated lines within a factor of 2 of the reported figure stays between 26% and 32% across every choice swept. Finer SIC levels (32% at 3 digits) do not repair it; the median intensity factor lowers the aggregate to 0.87 of reported.



Limits, and what I would build next

The assumption that breaks the conclusion

The book is simulated and the numerator is US direct emissions only. A multinational's balance sheet is global while its GHGRP emissions are the US facilities, so reported Scope 1 here is a floor for such borrowers and the attribution factor is not matched to the numerator. Scope 2 and 3 are absent. The size of the validated error is a property of this borrower set; a book with a different size mix will not reproduce it. That the reporters are the large firms is an assumption of the score rule, and the reported share is the choice that moves the result most (0.23).

What a sceptic attacks first

Loan amounts are random, so the 10.12 Mt describes no real bank. Matched parents cover 37% of reported emissions; the rest are private or foreign. Books overlap across the 300 draws, so the coverage rates are not independent trials.

Nothing here is forward-looking

The estimate is a snapshot of one reporting year; the held-out test above is across borrowers, not across time. The year sweep (2021-2023) is a robustness check.

Next

Fit intensity on log revenue within sector, which the size slope says is missing, and re-run the same calibration. Add Scope 2 from the same reports. Replace the simulated bank with a public disclosure of sector exposures when one exists at line level.